

Deep Learning-Based Relay for Online Fault Detection, Classification and Location in a Grid-Connected Microgrid

Introduction

The objective of this project is to develop an intelligent automated protection system that can quickly identify electrical faults occurring in a transmission line network.

Whenever a fault occurs in a power system, engineers need to know:

1. **What type of fault has occurred?**
2. **Where exactly has the fault occurred?**

Traditionally, identifying faults can be time-consuming and may lead to power interruptions. To solve this problem, we developed a Deep Learning-based smart relay system capable of automatically detecting, classifying, and locating faults with very high accuracy.

Dataset Used

The dataset was obtained from **IEEE Dataport** and contains:

- 578,923 electrical records
- 45 electrical parameters
- Voltage and current measurements from different transmission line sections
- Multiple fault conditions and fault locations

The dataset represents both normal operating conditions and various fault scenarios in a power transmission network.

Development Platform

The entire project was implemented using **Google Colab**.

Google Colab was chosen because:

- It provides cloud-based computing resources.
- It supports Python and Deep Learning libraries.
- It can efficiently handle large datasets.
- It allows model training without requiring high-end local hardware.

Libraries Used

Several Python libraries were used throughout the project:

Pandas

Used for loading, organizing, and managing the dataset.

NumPy

Used for numerical calculations and data transformation.

Matplotlib and Seaborn

Used for generating graphs, performance plots, and confusion matrices.

Scikit-Learn

Used for:

- Train-Test Split
- Data Standardization
- Label Encoding
- Model Evaluation

TensorFlow and Keras

Used to design, train, and evaluate the deep learning models.

Project Workflow

Step 1: Data Collection

The IEEE dataset containing voltage and current measurements was loaded into the Google Colab environment.

The dataset contains information about different fault conditions and transmission line segments.

Step 2: Data Preprocessing

The raw electrical data was cleaned and prepared for model training.

The preprocessing included:

- Checking for missing values
- Separating input features and target outputs
- Converting categorical labels into numerical form
- Standardizing electrical measurements

After preprocessing:

- 90% data was used for training
- 10% data was used for testing

This ensures that the model is evaluated on completely unseen data.

Step 3: Model Development

Two different deep learning models were developed and compared.

Existing Model (Baseline)

The first model uses a Bidirectional LSTM architecture.

This model learns patterns from electrical signals over time and predicts faults.

Proposed Model

The second model combines:

- CNN Layer
- Bidirectional LSTM Layer
- Attention Mechanism

This hybrid architecture allows the system to:

- Detect sudden abnormalities in electrical signals
- Learn signal behavior over time
- Focus on the most important fault-related information

Dual Output Prediction

The proposed model performs two tasks simultaneously:

Output 1

Fault Type Classification

Examples:

- No Fault
- SLG
- DLG
- L-L
- L-L-L
- L-L-L-G

Output 2

Fault Location Identification

Examples:

- L12
- L13
- L23

- L24
- L25
- L34
- L45

Thus, a single model can determine both the fault category and its exact location.

Training Optimization

To improve learning and avoid overfitting, the following techniques were used:

Batch Normalization

Improves training stability and speed.

Dropout

Prevents memorization of training data.

Adam Optimizer

Efficiently updates model parameters.

Early Stopping

Automatically stops training when further improvement is not observed.

Performance Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics help determine how effectively the system identifies different fault conditions.

Results

Fault Type Classification

Model	Accuracy
Existing LSTM	92%
Proposed CNN-LSTM	97%

Fault Location Identification

Model	Accuracy
Existing LSTM	96%
Proposed CNN-LSTM	98%

Key Observation

The traditional LSTM model struggled when dealing with complex and less frequent fault types.

The proposed CNN-LSTM model successfully identified both common and difficult faults with significantly higher precision and recall.

This demonstrates that combining CNN, Bi-LSTM, and Attention mechanisms improves the model's ability to learn complex electrical fault patterns.

Conclusion

This project successfully developed an intelligent Deep Learning-based relay system capable of automatically detecting, classifying, and locating faults in transmission line networks.

The proposed CNN-LSTM model achieved:

- 97% accuracy in fault type classification
- 98% accuracy in fault location identification

The results show that the system can provide fast, accurate, and reliable fault analysis, making it highly suitable for modern smart grid and automated power system protection applications.

1-Minute Summary for Faculty

"We developed a Deep Learning-based smart relay system using Google Colab and an IEEE Dataport dataset containing more than 5.7 lakh electrical records. After preprocessing and standardizing the data, we trained two models: a traditional Bidirectional LSTM model and a proposed CNN-BiLSTM-Attention model. The system simultaneously predicts the fault type and fault location in a transmission line network. The proposed model achieved 97% accuracy for fault classification and 98% accuracy for fault location identification, outperforming the existing approach and demonstrating its suitability for real-time smart grid protection."